Before the
New York State Public Service Commission

In the Matter of

Consolidated Edison Company of New York, Inc.

Case 25-E-0072
Case 25-G-0073

May 2025

Prepared Direct Testimony of:

New York City Electric
Infrastructure Panel

Joseph Martorano
Charlotte Dean
Odibert Eid
Daniel Guralchuk
Chandler Stewart
Gabrielle Savona

On Behalf of:

The City of New York

INTRODUCTION

Q. PLEASE STATE YOUR NAME, OCCUPATION, AND BUSINESS ADDRESS.

A. Mr. Martorano – My name is Joseph Martorano P.E. I am an associate principal at Arup overseeing the energy disciple in the east geography of the U.S. My business address is 1085 Raymond Boulevard, Suite 1800, Newark, NJ 07102.

Ms. Dean – My name is Charlotte Dean, PE. I am a Senior Engineer at Arup. My business address is 77 Water Street, NY, NY 10005.

Mr. Eid – My name is Odibert Eid. I am a consultant at Arup. My business address is 77 Water Street NY, NY 10005.

Mr. Guralchuk – My Name is Daniel Guralchuk, PE. I am an associate principal at Arup and functioning as the Energy Business Leader within the Eastern Regional US. My business address is 1085 Raymond Boulevard, Suite 1800, Newark, NJ 07102.

Mr. Stewart – My name is Chandler Stewart. I am an Engineer at Arup. I am a part of the Power Systems team. My business address is 1120 Connecticut Ave NW, Suite 1110, Washington, DC 20036.

Ms. Savona – My name is Gabrielle Berns Savona, PMP. I am a consultant on the Energy Team in Arup's Boston office. My business address is 60 State Steet in Boston, Massachusetts 02109.

Q. ON WHOSE BEHALF ARE YOU SUBMITTING THIS TESTIMONY?

A. The City of New York, or City.

Q. PLEASE DESCRIBE YOUR EDUCATIONAL BACKGROUND.

A. Mr. Martorano – I received a Bachelor of Science degree in Mechanical Engineering from The Pennsylvania State University in 1985.

Ms. Dean – I received a Bachelor of Science degree in Civil Engineering and Engineering Mechanics from Columbia University in 2007 and a Master of Science in Mechanical Engineering from the University of Pennsylvania in 2013.

Mr. Eid – I received a Bachelor of Science degree in Electrical Engineering from Hofstra University in 2022.

Mr. Guralchuk – I received a Bachelor of Science degree from Villanova University in Electrical Engineering in 2000 and a Master of Business Administration from Baruch College in 2008.

Mr. Stewart – I received a Bachelor of Science degree in Electrical Engineering from the University of Maine in 2018.

Ms. Savona – I received a Bachelor of Science degree from the University of Massachusetts, Amherst in Chemical Engineering and a minor in Economics in 2019.

Q. PLEASE DESCRIBE YOUR PROFESSIONAL EXPERIENCE.

A. Mr. Martorano – I have 40 years' experience in the energy industry. I spent 11 years with Exelon Corporation where I served as the Director of Pricing. I have worked on the wholesale side of the energy industry managing the supply portfolio of various clients. I have held the position of COO of SOSH Architects

from 2005 to 2008, General Manager of Veolia Mid-Atlantic for 3 years, Senior Vice President of Kiely Engineering from 2017 to 2021, Executive Vice President of Greener by Design from 2008 to 2015, Vice President of Northeast-Western Energy for 2 years and President of Economics Energy. I am a licensed professional engineer in New Jersey, Pennsylvania, Delaware, Maryland and Washington, D.C.

Ms. Dean – My past work includes over 15 years' experience in district scale energy plants including design, construction administration, feasibility studies, master planning, technical due diligence, business development, regulatory compliance, utility billing, project management and asset management. I am a Registered Professional Engineer in New York and California. I am also a LEED accredited professional, a member of the American Society of Heating, Refrigeration and Air Conditioning Engineers, and a member of Urban Green/US Green Building Council New York Chapter.

Mr. Eid – My experience began with a focus on large-scale battery storage systems and electric vehicle charging infrastructure during a dedicated internship with Rhynland Energy. For the past two years, I have worked at Arup as an Energy Engineer. My work involves technical reviews for renewable energy projects, planning for EV charging solutions, complex energy modeling for microgrids and utility systems, and helping develop decarbonization plans for various clients.

Mr. Stewart – I have worked in the energy industry for six years. I spent three years of my career at Mott Macdonald, where I focused on physical substation

design. My past three years at Arup have included similar responsibilities with the addition of performing interconnection and power systems studies and preparing interconnection applications. I work primarily in the New England Independent System Operator territory, the New York Independent System Operator, Inc. territory, and the Pennsylvania-New Jersey-Maryland Interconnection territory.

Mr. Guralchuk – I have over 25 years of experience in the power and energy business sector with particular focus on electrical design, project/program management, and construction management related to electric power transmission and distribution systems, substations, and critical control facilities. I am currently an Associate Principal and have been serving as the Energy Business Leader for Arup since September 2024. In the past, I have held various positions at other firms such as Burns Engineering serving as a Senior Project Manager from December 2019 to August 2024, Enercon Services serving as Senior Consultant from September 2015 to April 2019, AECOM serving as a Project Manager from November 2008 to September 2015, Consolidated Edison Company of New York, Inc. serving as a Project Engineer from August 2004 to October 2008, and PSEG Power serving as an Electrical Engineer from July 2000 to August 2004.

Ms. Savona – I have nearly six years of experience in the energy industry. I started my career in oil and gas risk engineering at ExxonMobil in Houston, Texas from August 2019 to January 2021. I then began work in solar development and renewable energy generation working as a solar and battery

energy storage system developer outside of Boston, Massachusetts from February 2021 to July 2022. I have been with Arup for approximately three years – since July 2022 – focusing on project management and energy engineering. My work has spanned diverse sectors—including transportation, public and private development, aviation, and the broader energy industry—providing me with a comprehensive understanding of how generating assets are implemented and how they interact with and impact the electrical grid.

Q. HAVE YOU EVER TESTIFIED BEFORE ANY AGENCY OR COURT OR IN ANY OTHER VENUE ON THE TOPICS YOU ADDRESS IN THIS TESTIMONY?

A. Mr. Martorano – Yes. I prepared and provided testimony related to the Limerick Nuclear power plant entering rate base in 1989/1990, on behalf of PECO Energy in several rate cases in the 1990's when I was Director of Pricing at PECO Energy. I also testified before the New Jersey Senate in 2013 on matters related to resilience and reliability post Super Storm Sandy. I also prepared written testimony for the then Mayor of Hoboken, Dawn Zimmer, in 2013 when she appeared before the U.S. House of Representative subcommittee reviewing responses to Super Storm Sandy.

Ms. Dean – No.

Mr. Eid – No.

Mr. Guralchuk – No.

Mr. Stewart – No.

Ms. Savona – No.

PURPOSE OF TESTIMONY

Q. WHAT IS THE PURPOSE OF YOUR TESTIMONY?

A. Our testimony will cover three areas pertaining to Con Edison's electric rate filing. We comment on the testimony and associated exhibits, supplements, and updates for Con Edison's Electric Infrastructure and Operations Panel, or EIOP, the Company's Information Technology Panel, or IT Panel, and the Company's Shared Services Panel, or SSP. Our technical review of Con Edison's filings is limited to determining if the proposed projects and costs and allocations per year, in our opinion, are prudent and properly documented.

While we recognize that the Company is not proposing a multi-year rate plan, our testimony will discuss the Company's Rate Year ("RY1") capital plant additions in the context of the Company's recent historical investments, and its proposed investments in the two subsequent years ("RY2" and "RY3") following the Rate Year. To consider these investments solely in the context of the Rate Year is not meaningful because many of the Company's investments have been underway for several years and are forecasted to span multiple years or, in some cases, decades.

Q. ARE YOU SPONSORING ANY EXHIBITS WITH YOUR TESTIMONY?

A. Yes. We offer Exhibit__(NYCEIP-1), which is a compilation of the discovery responses from Con Edison that we relied upon in preparing our testimony, some of which we expressly reference.

ELECTRIC INFRASTRUCTURE AND OPERATIONS

Q. WHAT ARE YOUR GENERAL OBSERVATIONS REGARDING CON EDISON'S PROPOSED INVESTMENTS AS DESCRIBED IN THE COMPANY'S ELECTRIC INFRASTRUCTURE AND OPERATIONS PANEL DIRECT TESTIMONY?

A. The City fully appreciates the reality that Con Edison's electric infrastructure is aging; and that over the next 10 years, there will be a need for capital investment throughout New York City and the Company's service territory. However, while Con Edison appears to be fulfilling its obligation to provide reliability and resiliency, there appears to be inconsistencies with some programs and their budgeting. A fair number of programs included in Con Edison's capital forecasts are ongoing, yet in this case the Company is requesting larger capital budgets for such projects as compared to forecasts and actual project spending. While year-over-year increases are expected, in some cases, the amount of increased capital is significant. The projects and specific concerns highlighted in this testimony represent the most significant examples of where this is occurring without appropriate justification.

Furthermore, we believe that in some instances, the Company's timing for implementation of capital and staffing projects may be too accelerated and would better serve ratepayers if postponed or delayed. In other instances, some of the Company's approaches to capital projects assume scenarios that do not properly represent what is the most likely and the worst-case scenarios. Generally, Con Edison's analyses are based on a future scenario that do not fully

reflect the current state of the energy sector, load growth, and economic conditions relevant in the near term.

Additionally, some large-scale projects hinge upon other prerequisites that are not fully discussed. For example, some projects have a prerequisite for large-scale offshore wind interconnection into the Con Edison system. Given the present state of offshore wind and the canceling or delay of leases and projects, significant capital outlays by Con Edison at this time to meet this growth may not be warranted and could be postponed. Accordingly, we question the validity of the timeframes of these projects.

Finally, several projects rely on the underlying premise that supply chain issues that began during the pandemic are still of concern. These issues have become less of a bottleneck, and what was prudent for procurement at that time may not be as prudent now. Alterations to the timing and scope on several projects due to the above factors will reduce the necessary revenue requirement in 2026, while still providing ratepayers with reliable and resilient electric service with enough capacity to meet growth forecasts.

Q. WHAT IS THE COMPANY PROPOSING FOR ITS TRANSFORMER PURCHASE AND REPLACEMENT PROGRAMS?

A. The Company is proposing a suite of programs targeted at transformer procurement and replacement, totaling over $660 million. This includes the proactive "Substation Transformer Replacement Program" discussed in Exhibit__(EIOP-8) at page 268 totaling $508 million through RY1, RY2, and RY3. The Company has also allocated $138 million for spares due to cited five

to six year lead times. The Company's "Failed Substation Transformer Program" discussed on page 6 of Exhibit__(EIOP-9) is based on equipment that has approximately 3 failures per year with a cost of $153 million over RY1, RY2, and RY3. The Company also proposes the "Transformer Purchases" program and "Network Transformer Relief" program within its $660 million budget.

Q. DO YOU HAVE ANY CONCERNS?

A. Yes. Our primary concern is the lack of an integrated, fully justified risk-based strategy connecting these significant expenditures. Specifically:

- Lead Times versus Reactive Budget: If transformer procurement lead times have indeed extended to five to six years justifying a $138 million investment in long-lead time spares, then budgeting the reactive replacement program based on only approximately three failures per year may underestimate the system risk and potential impacts should unexpected failures occur before spares are available.
- Obsolescence and Stranded Assets: Procuring specialized, high-value assets five to six years in advance introduces risks of technological obsolescence, changes in system requirements, or stranded assets if forecasts or needs change.
- Cost-Effectiveness: The Company has not demonstrated that this combined strategy is the most cost-effective method for managing transformer failure risk compared to clearly defined alternatives.

- Transformer Purchases Program Specifics: This program is meant to provide either opportunistic replacement of 500 kVA transformers or to purchase transformers for future new business requests. It states an intent to replace 500 kVA transformers with 750 kVA units, the highest capacity option fitting existing structures. While extra headroom is generally good, the program does not seem to be selective for transformers actually near their 500 kVA threshold. In cases where load forecasting indicates demand would not exceed 500 kVA, replacement with another 500 kVA unit would be more cost-effective than a more expensive 750 kVA unit. Additionally, the program mentions future purchasing of dry-type submersible underground distribution transformers without detailing the cost difference, reliability, or maintenance needs compared to existing infrastructure. A review of existing infrastructure proposed new transformers, and the differences between the two is needed so that Con Edison is making the most prudent and cost effective investment in its system.
- Network Transformer Relief Program Budget: For this project, there's a large discrepancy between 2025 expenditures and costs proposed for the Rate Years. The 2026 total capital budget is 2.37 times the 2025 budget, while the number of transformers addressed is only 1.088 times greater. This causes a unit cost per generic transformer on average to increase from about $88,000 per unit to over $193,000 per unit, with no

explanation for this marked increase. Assuming 2025 costs for the remaining rate case years could greatly reduce this project's cost.

- Failed Substation Transformer Program Budget: This project requests $239 million. A significant portion of the cost is split equally (*e.g.*, $51 million per year for 4 years in one analysis). While the total value may be justified, this methodology for allocating the capital request over the project term seems arbitrary and inappropriate. A more detailed breakdown and accounting of costs by year is necessary.

Q. HOW CAN YOUR CONCERNS WITH THE COMPANY'S TRANSFORMER PURCHASE AND REPLACEMENT PROGRAMS BE ADDRESSED?

A. To assess the Company's overall strategy for transformer purchases, the Commission should direct Con Edison to provide the following additional data:

- Verifiable documentation for the asserted five-to-six-year procurement lead times.
- A specific risk analysis demonstrating that assuming approximately three failures per year for the reactive replacement budget will adequately provide for system reliability during the spare replenishment lead time.
- A detailed explanation of the strategy to mitigate obsolescence and mismatch risks for transformers procured far in advance.
- A comparative analysis justifying the cost-effectiveness of the proposed combined investment approach versus alternatives.

- A detailed justification for the unit cost increases in the Network Transformer Relief program and the annual budget allocation for the Failed Substation Transformer Program.
- A more selective approach for the Transformer Purchases program based on an actual need for uprating.

Q. PLEASE DISCUSS THE BROOKLYN CLEAN ENERGY HUB SCALABLE RELIABILITY PROPOSAL AND RELATED PROJECTS LIKE BROOKLYN ENERGY HUB (POSITIONS 2S AND 2N).

A. The Brooklyn Clean Energy Hub, or BCEH project, with a capital request of approximately $633 million ($162 million targeted for 2026), aims to facilitate the interconnection of offshore wind and provide electric supply for the Brownsville and Gateway substations. A related project, the Brooklyn Energy Hub (Positions 2S and 2N), calls for the addition of two 345 kV positions within the BCEH.

Q. DO YOU HAVE ANY CONCERNS?

A. Yes. Our concern is that these expenditures, particularly the $162 million for the BCEH in 2026, appear excessive and untimely given the present state of offshore wind development into New York City. While we support the offshore wind program and submit it is a necessary tool in the decarbonization of New York, we believe that the timing as outlined in the Company's filing is not warranted. Known offshore wind projects have experienced delays and there is added hesitancy within the market. Further, there have yet to be any posted project Construction and Operations Plan submissions that identify the

Brooklyn Clean Energy Hub as a point of interconnection. An overwhelmingly large percentage of the projects that the BCEH is being built to support are either significantly delayed or have been cancelled.

Based on a study by Aurora Energy research, 32 GW of the planned Atlantic offshore wind projects are at high risk of termination. While the administration and New York have agreed to move forward with the Empire Wind 1 wind project, this is only one of many such projects that may be in jeopardy. Combined with this market hesitancy, spending substantial ratepayer funds for something that cannot be fully utilized until a much later date would not be prudent at this juncture given the magnitude of the Company's rate increase request.

Con Edison, in its white paper for the 2S and 2N positions in Exhibit__(EIOP-7), discusses an alternative: Alternative 1: Add the two additional 345kV circuit breakers after the station has been energized. The Company rejected this due to challenges in securing equipment outages. However, given the marketplace and offshore wind implementation realities, the delays and issues raised as justifications for this alternative being non-viable may no longer be valid. Pursuing this alternative would be more prudent, allowing for flexibility if changes or further timing adjustments occur.

Similarly, for the broader BCEH, Con Edison presented a reduced scope alternative for building it (single ring bus instead of a double ring bus), noting it could be completed in time for substation supply but would cost more to expand later.

Q. WHAT IS YOUR RECOMMENDATION?

A. Due to current market uncertainty and offshore wind development delays, the full build-out of the BCEH is likely not needed at this time. It would be best to revisit the alternatives proposed by the Company in its testimony to better understand at what point a second ring or full capacity would genuinely be necessary. Based on our review of page 18 through 26 of Exhibit__(EIOP-7), the Company's white paper for the BCEH appears to have been created without adjustment for these significant market changes. Accordingly, we recommend that the Commission reject the capital investment as proposed and require the Company to revisit its alternative options.

Q. DO YOU HAVE ANY CONCERNS WITH THE COMPANY'S AREA SUBSTATION PHASED REPLACEMENT PROGRAM?

A. Yes. This program aims to replace medium voltage equipment and associated civil work at various substations. It is an ongoing program, but it shows a significant rise in requested capital. In 2026, an estimated need of $33 million is projected. While this magnitude seems consistent with their following years of planning from 2027 to 2029, it is approximately three times as much capital as the 2024 forecast and the 2025 requested amount. A majority of the 2026 to 2029 forecasted period appears to be focused on three stations: East 63rd St., East 75th St., and Plymouth Street.

The program noted engineering and procurement began in 2022, with equipment purchases scheduled for 2025. While cost increases associated with vendor pricing are expected, this is an ongoing program with a constant rotation

of new substations. We would expect the capital costs of these replacement programs—future and past—to be along the same level of magnitude for capital needs, not such a sharp increase. No explanation was provided for this change in funding. Accordingly, we recommend that Commission only allocate funding based on the Company's historic spending levels.

Q. DO YOU HAVE ANY CONCERNS WITH THE AUXILIARY STATION EQUIPMENT REPLACEMENT PROGRAM?

A. Yes. This ongoing program, used since at least 2019, aims to upgrade essential auxiliary equipment in substations. Historically, the program costs and budget were approximately $1 million to $2 million per year. The program now requests an increase to at least $13 million each year. This level of increase is significant - greater than six times the historical amount.

The white paper on page 13 of Exhibit__(EIOP-8) states: "[t]his program has been expanded to encompass additional work, which drives the increased funding request for 2026 and beyond." However, it does not provide sufficient justification or a detailed breakdown of the additional work for such a substantial increase, especially when actual spending costs were relatively close to the historically budgeted amounts. Accordingly, we recommend that the requested increase in budget for this project should be rejected until further evidence justifying this spend is presented.

Q. WHAT IS THE COMPANY'S PROPOSAL FOR ITS PRIMARY FEEDER RELIABILITY PROGRAM?

A. The Primary Feeder Reliability Program on page 189 of Exhibit__(EIOP-8) requests approximately $65.5 million in RY1, $67 million in RY2, and $66 million in RY3, totaling $198.5 million. This program plans to remove paper insulated lead covered cable, install new interrupters, and split and extend feeders. This Program is an ongoing initiative. However, the Company also provides two alternatives. These are:

- Alternative 1: Replace only the high failure rate transition splices with a newer, more reliable splice design. This would reduce program costs by one-third.
- Alternative 2: A cable diagnostic system to accurately determine paper insulate lead cable, or PILC health, targeting only unhealthy cable for replacement.

Q. WHAT ARE YOUR THOUGHTS ON THESE ALTERNATIVES?

A. While Con Edison notes these alternatives might be less reliable or that diagnostic systems have not proven effective, implementing them could provide needed relief at a fraction of the cost, potentially remedying the problem or at least postponing substantial costs to ratepayers. The cost for these alternatives is not listed so the cost differential between the Company's proposal versus the alternates is not known, but based on scope could be significant. If these alternatives provide a permanent solution, their temporary relief will push added costs out several years. We recommend that the Company be required to explore the use of such alternatives.

Q. WHAT IS THE COMPANY'S PROPOSAL FOR THE UNDERGROUND SECONDARY RELIABILITY PROGRAM?

A. The Underground Secondary Reliability Program on page 295 of Exhibit__(EIOP-8) requests approximately $35.5 million in RY1, $37 million in RY2, and $38 million in RY3, totaling $110.5 million. This program is ongoing and would be used to replace and upgrade underground secondary equipment.

Q. DO YOU HAVE ANY CONCERNS WITH THIS PROGRAM?

A. Yes. The information provided indicates that in 2025, Con Edison is projecting 11,000 linear feet to be serviced at an estimated cost of $7 million. In the rate case years, the cost appears to be roughly 5 times the 2025 total for approximately 72% more feet of cable. This would be an excessive increase in cost per foot and is not supported by any justification within the whitepaper.

Q. WHAT ARE YOUR RECOMMENDATIONS?

A. To adequately assess this program, the Commission should direct Con Edison to provide:

- A quantitative forecast (RY1-RY3) for new, repaired, AI-identified, and net year-end backlog of total open mains.
- An analysis of funding/operational plan required to hold the total backlog flat and to reduce it by 10% annually.
- A quantitative risk assessment for the growing backlog of Priority 3-5 and AI-identified mains.

- Justification for the significant cost-per-foot increase in the "Underground Secondary Reliability" project compared to 2025.

Pending the outcome of these additional analyses, the Commission should consider if the full $110.5 million is justified or if a revised budget linked to demonstrably controlling the total backlog is warranted.

Q. PLEASE DISCUSS THE AGGREGATE JUSTIFICATION FOR BROAD RELIABILITY PROGRAMS LIKE NON-NETWORK RELIABILITY, PRIMARY FEEDER RELIABILITY, AND UNDERGROUND SECONDARY RELIABILITY AS PRESENTED IN EXHIBIT__(EIOP-8).

A. The Company's reliability programs found in Exhibit EIOP-8 include the Non-Network Reliability program, with a $239.4 million budget for RY1 through RY3, the Primary Feeder Reliability program with a $198.0 million budget RY1 through RY3, and the Underground Secondary Reliability program with a budget of $110.5 million for RY1 through RY3, collectively totaling approximately $547.9 million. The Company generally justifies these programs by targeting poorly performing assets to improve reliability metrics.

Q. DO YOU HAVE ANY OBSERVATIONS ABOUT THESE PROGRAMS?

A. While the goal of improving poorly performing assets is fundamental, the Company's supporting project exhibits often lack the specific, granular detail to assess the prudence and targeted effectiveness of this nearly $550 million spending. Specifically:

- It's difficult to determine the precise types of projects and equipment interventions constituting the bulk of spending within each distinct program.
- The quantitative criteria differentiating work allocated to these proactive Exhibit__(EIOP-8) programs versus related reactive replacement programs in Exhibit__(EIOP-9) (*e.g.*, Primary Cable Replacement, Secondary Open Mains) are not clearly defined, raising potential for overlap or inefficient allocation.
- The Company's exhibits lack specific, measurable reliability improvement targets directly and exclusively attributable to the requested funding level for each individual large program.

Q. DO YOU HAVE ANY OTHER CONCERNS?

A. Yes. For the Non-Network Resiliency Program – which appears to be the same as Non-Network Reliability Programs – there is a specific budgeting concern. This project uses a formula in tables, as seen on page 29 of Exhibit__(EIOP-10), referring to 2023 values as a baseline. When reviewing this table, it appears that based on the 2023 values, the outlined figures for 2026 are off by about $3 million. Con Edison should review these calculations.

To ensure these funds are targeted effectively, the Commission should direct Con Edison to provide for each program:

- A detailed breakdown (RY1-RY3) of specific capital projects/equipment interventions and budget allocation per type.

- Explicit quantitative criteria for selecting assets under this program, ensuring no duplication with Exhibit__(EIOP-9) reactive programs.
- Specific, measurable, and targeted annual improvement forecast as a direct result of the proposed funding for that individual program.

Q. ARE THERE CONCERNS WITH STAFFING-RELATED BUDGET REQUESTS?

A. Yes, for Central Engineering Staffing and Position Additions, the project appears to show a 14.9% increase in cost from historical values as presented in the filed white paper of Exhibit__(EIOP-5). There does not appear to be a substantial difference in the program from past years, and this increase appears unwarranted. A more modest increase due to inflation would be more reasonable.

Furthermore, based on the assessment that several capital projects may not be needed for deployment in 2026 – particularly those linked to delayed offshore wind development – it would make sense that staffing needs associated with managing these projects would also be reduced. Accordingly, the project budget as proposed is not justified and should be modified.

Q. DO YOU HAVE CONCERNS ABOUT PROJECTS RELATED TO THE ATLANTIC AREA SUBSTATION?

A. Yes. For the Atlantic Area Substation project, it is anticipated that this substation would be supplied power from the planned Brooklyn Clean Energy Hub. As we previously discussed, this Hub may not be able to support providing power in the outlined timeframe as its major sources of generation are offshore

wind and a majority of those projects have been significantly delayed or cancelled. This project may make sense at a later point in time, but with the current state of offshore wind development, it would be a risk to ratepayers to expend money until there is certainty about generation.

Q. WHAT IS YOUR RECOMMENDATION?

A. Similar to our recommendation for the Brooklyn Clean Energy Hub Project, this project as proposed should not be included in the capital budget for this rate proceeding as it is premature due to the uncertainty of offshore wind generation project development.

Q. WHAT IS THE COMPANY'S PROPOSAL FOR THE BROWNSVILLE AREA LOAD RELIEF PROJECTS?

A. Several projects are linked to load relief in the Brownsville area, predicated on revised forecasts assuming significant EV load growth and electric heat adoption. Such projects include the Brownsville Area Load Relief – Establish Gateway Park No. 1, Brownsville Area Load Relief – Glendale TR5 and Feeder 38Q05 Extension, and the Crown Heights Network Split - Brownsville No. 1 to Gateway Park No. 1.

In Exhibit__(EIOP-7), the white paper for Gateway Park No. 1 outlines load relief actions providing a cushion until 2027 under the forecasted load. However, as evidenced by industry studies, EV adoption rates have slowed compared to the Company's forecasts. Projections in early 2024 anticipated EVs would comprise approximately 12% of new vehicle sales; by mid-2024, this was revised downward to approximately 9%. J.D. Power projects EVs will

maintain a 9.1% share through 2025, indicating a plateau. This slowdown makes the projected overcapacity concerns less immediate.

If this 3% reduction in new EV load holds, the 2027 capacity issue could be moved to 2029. This would make the timing of these projects premature, allowing for delays in spending in 2026 and 2027. Delaying would also allow Con Edison to better re-evaluate market conditions. The justifications for the Glendale TR5 project and the Crown Heights Network Split mirror these assumptions and therefore face the same issues regarding timing. These projects should be delayed until a better understanding of growth is established, with alternative relief efforts taken in the interim.

Importantly, we recommend that taking the needed steps to address existing issues related to resiliency and reliability of existing load should move forward, and the City supports investments necessary for companies and warehouses that have *actual* plans for fleet transition in these neighborhoods, which will provide localized air quality benefits. Aside from concrete fleet transition plans of area businesses, work associated with or required due to the Company's unsupported and generalized load growth forecasts should be delayed based on what we have outlined above.

Q. WHAT IS THE COMPANY'S PROPOSAL FOR THE PARKCHESTER AREA PROJECTS?

A. The Company proposes three projects for the Parkchester area. These include the Parkchester No. 1 5th Transformer 9S and Supply Feeder 38X05, Parkchester No. 2 TR13 & B/S 13A & 13B Installation, and Parkchester No. 2

Upgrade Limiting Bus Sections project. Each of these projects reference the same justification for project need, which is that the capacity of serving systems will be exceeded which can be found in Exhibit__(EIOP-7).

Q. DO YOU HAVE ANY CONCERNS?

A. Yes. Based on the data provided by Con Edison in Exhibit__(EIOP-7), exceedance of capacity does not occur until 2030. The data also shows load adjustments that peak at 16 MW in 2028. The white paper also shows that this load relief of 16 MW is discontinued after 2028. There is no apparent reason given why this peak load reduction does not continue to be available for subsequent years; in fact, the white paper calls for it to reduce in subsequent years. If this peak load reduction holds steady for several more years, it is our opinion that the Parkchester Area projects could be delayed by up to two years, deferring project costs in 2026 through 2028. This also assumes load growth continues at its present pace without any significant economic downturn.

Q. **WHAT IS THE COMPANY PROPOSING FOR THE WASHINGTON STREET TO CEDAR STREET TRANSFER PROJECT?**

A. Based on the Company's latest 2024 Forecast, the Washington St. Area Substation requires a load transfer in 2031 to mitigate risk. As a result, a proposed load transfer of approximately 50 MW will be transferred from Washington St. to Cedar St. Area Substation

Q. DO YOU HAVE CONCERNS?

A. Yes. Similar to the Parkchester projects, here, based on the data presented by the Company, capacity is not exceeded until 2032. The need for transferring

load and the amount of transferred load may not justify the size and scope of this project. The whitepaper does not present a compelling argument for why load reduction actions could not alleviate overcapacity for as many as 3 years, which would decrease the capital budget and provide ratepayer relief. In addition, the expenditure for 2026 seems high, as it is approximately one third of the project costs while only 20% of work is being completed.

Q. DO YOU HAVE ANY CONCERNS WITH THE COMPANY'S NEW BUSINESS CAPITAL BUDGET?

A. Yes. While there is a strong history of this line item's value, the issue we have concerns with is the scale of change proposed as compared to historical spending. Actual costs from 2020 to 2024 show escalation per year at a rate of approximately 6%. Between 2024 and 2025 there is step change in expenditure with an approximately 40% increase. This increase then flattens out. On average, there has been a 6% year-over-year levelized increase from 2020 to 2024. That percentage is called for in 2026 and 2027, but the initial large step change of 40% seems unwarranted and should be adjusted to better represent history and future growth.

Q. WHAT IS THE COMPANY'S PROPOSAL FOR THE BUSHING REPLACEMENT PROGRAM?

A. A bushing is a device that brings out the transformer internal winding leads through an insulated tube for connection to the power system. Utility industry experts have identified transformer bushings that have had design and manufacturing problems. The identified bushings are Pfisterer Connex plug-in

bushings and General Electric Type U bushings. Failure of the bushings can lead to transformer failures, decreased system reliability and availability of transformers on the transmission and sub transmission systems.

Approximately thirty 345 kV bushings, two-hundred and forty 138 kV bushings, twenty-five 69 kV bushings, and ten 23 kV bushings have been identified for replacement and upgrade. As of 2024, approximately ninety bushings remain on the system to be replaced. It is recommended that identified bushings be replaced and upgraded based on failure probability and system impact. Work has already been completed on all 138kV and 345 kV shunt reactors, 345 kV transmission autotransformers, and phase angle regulators (PARs). The recommended priority for work remaining is: (1) Other 138 kV and 345 kV auto-transformers, (2) Two bank area substation transformers, and (3) Other area substation transformers.

Q. WHAT ARE YOUR CONCERNS?

A. Similar to the Auxiliary Station Equipment Replacement Program, the budget for the Bushing Replacement Program increases by approximately three-fold from original values and historical spending. There is not enough justification within the white paper provided by Con Edison to approve such a significant increase. We recommend that more analysis and supporting data is needed from the Company before such a large increase to this project's budget spending can be approved.

Q. WHAT IS THE COMPANY PROPOSING FOR THE E. 179TH ST. FEEDER ESTABLISHMENT PROJECT?

A. This project will enhance the reliability and capacity of the Fordham network by establishing ten new 13 kV primary feeders at the E. 179 St. area substation. The new feeders will leverage the feeder positions constructed during the E. 179 St. area substation rebuild. The primary objective is to lower the Network Reliability Index, or NRI, for the Fordham network, which is currently ranked eleventh in network reliability within the Con Edison service territory. The new feeders will also address the issue of heavily loaded existing feeders, which are operating above 90% of their normal and emergency ratings.

Q. WHAT ARE YOUR OBSERVATIONS AND DO YOU HAVE ANY CONCERNS?

A. Generally, a review of the Company's workpapers in Exhibit__(EIOP-8) appear to support the need for this project as a means for providing reliable service and its overall cost seems properly budgeted. However, we have concerns with the timeline and schedule for the project. While the aggressive nature of the schedule is commendable, it may be unrealistic given procurement, engineering design requirements, and scope. The project would be better served if the start of construction (civil and electric) were pushed back until Q4 2026. This would allow more time for proper design, site survey, and determination of needed changes due to supply chain and other fluctuating market conditions. The 2024-2033 Area Substations and Sub-transmission Feeders Ten-Year Relief program shows these feeders may be at 95% capacity by 2027, which we believe allows for a several-month delay into 2027 to better assess needs and improve design.

Q. DO YOU HAVE CONCERNS WITH THE FIRE SUPPRESSION SYSTEMS UPGRADES PROGRAM AND HIGH VOLTAGE CIRCUIT BREAKER CAPITAL UPGRADE PROGRAM BUDGETING METHODOLOGY?

A. Yes. Both projects share the same issue with budgeting. The Company requests $42 million and $92 million respectively for a combined $134 million budget. While the total value may be justified and we take no position on the level of funding requested, the methodology the Company used for allocating the capital request over the four year term appears inappropriate and unsupported. Based on Exhibit__(EIOP-8), it appears that Con Edison simply divided the total budget request by four and spread the cost equally among the two projects for each year. A more detailed and thorough breakdown and accounting of costs by year should be performed for such a significant investment. Until such accounting is performed, this capital request by the Company does not appear reasonable.

Q. WHAT IS THE COMPANY PROPOSING FOR FUNDING FOR ITS TRANSFORMER VAULT AND STRUCTURES MODERNIZATION PROJECT?

A. This program provides funding for proactive repair of structural deficiencies in deteriorated transformer vaults, manholes and service boxes. If not addressed, structural deficiencies in deteriorated vaults present a risk of collapse that can be a hazard to the public and can compromise system reliability by causing damage to electric infrastructure.

Q. DO YOU HAVE CONCERNS?

A. Yes. Based on the schedule and white paper in EIOP – 8 the funding for this project is not justified and supported in any year as presented. While the project addresses a valid need, allocating 25% of the budget in the first year would be excessive given that most of the work will be design and base construction.

Q. WHAT ARE YOUR OBSERVATIONS FOR THE IDLEWILD 27KV AREA SUBSTATION PROJECT?

A. This Idlewild 27kV Area Substation project will transfer load from the existing Jamaica network to the new Idlewild substation, which will handle an electric load of 170 MW starting in 2028, and growing by 8MW to 178MW by 2034. This new substation calls for four 93MVA rated transformers with five sections of 27kV rated switchgear. If the total load is 178 MW by 2034, and if the four transformers equally shared this load - 44.5 MW each - then each transformer and its associated 27kV switchgear would have more than 50% spare capacity. Exhibit__(EIOP-7) states that spare capacity could be used for New York State's clean energy future goals, but it is difficult to justify that these goals would eclipse the initial load relief value without supporting evidence.

Q. WHAT IS YOUR RECOMMENDATION?

A. Based on our review of the supporting white paper it appears that Con Edison may be overbuilding the required substation which has a total capital cost of $355,067,000. We recommend that Con Edison review its required substation equipment for cost-effectiveness and plan for the appropriate amount of equipment needed in the identified timelines. Future expansion can still be an

option, but providing more equipment than is realistically needed is not an appropriate use of funds.

Q. DID YOU REVIEW THE PROPOSAL FOR A NEW ENERGY CONTROL CENTER?

A. Yes. The Company is proposing a very expensive project to build a new Energy Control Center valued at a total capital cost of $980 million. The Company presents this project as intended to meet the latest industry standards for physical and cyber security and utilize the latest technology. While the location of the new control center has not yet been determined, this new facility could potentially be placed anywhere in Con Edison's service territory.

Q. DO YOU HAVE ANY CONCERNS?

A. Yes. Recognizing that the new control center could be located outside of New York City, we have concerns that such a move from the Company's headquarters would introduce additional costs for personnel relocations and facility operation/maintenance. Furthermore, the replacement proposal does not demonstrate how response times and mitigation scenarios will be impacted or improved by this new center. The testimony also does not demonstrate if the existing Energy Control Center could be refurbished to meet industry requirements, utilizing the Alternate Energy Control Center during any transition.

Q. WHAT IS YOUR RECOMMENDATION?

A. Before making the determination that a Con Edison needs to build a completely new control center, an in-depth review of cost-effective solutions for replacing

the Energy Control Center and plan for appropriate infrastructure needed in the identified timelines should be performed. As mentioned earlier, any associated staffing increases should also be scrutinized, especially if underlying capital projects are delayed. Accordingly, the Company has not demonstrated that more cost-effective options do not accomplish the intended purpose and need of their proposed project. Before moving forward on a nearly billion dollar project, such diligence is needed.

Q. WHAT IS YOUR SUMMARY OF GENERAL CHANGES OR ACTIONS THAT YOU RECOMMEND THE COMPANY TAKE BASED ON YOUR REVIEW OF CON EDISON'S ELECTRIC INFRASTRUCTURE AND OPERATIONS TESTIMONY AND EXHIBITS?

A. Based on our review, and as a supplement to the recommendations already provided for specific projects, we recommend Con Edison:

- Re-evaluate the timing and necessity of capital projects and associated staffing linked to potentially delayed prerequisites, such as large-scale offshore wind development and overly optimistic near-term electrification adoption rates.
- Provide more detailed justifications for significant budget increases in ongoing programs compared to historical spending, particularly for unit cost escalations and large step-changes in annual funding. This includes programs like Auxiliary Station Equipment Replacement, Bushing Replacement, New Business Capital, and various transformer and cable programs.

- Adopt a more phased or flexible approach for large new infrastructure projects where demand is uncertain, considering alternatives that allow for scaling up as needs become more concrete (*e.g.*, Brooklyn Clean Energy Hub, Idlewild Substation).
- Ensure that budgeting for multi-year programs reflects a realistic projection of annual work and expenditure rather than simply dividing total costs equally over the period, as seen in programs like Fire Suppression Systems Upgrades, High Voltage Circuit Breaker Upgrades, and the Failed Substation Transformer Program.
- Provide clearer, quantitative justifications for how programmatic spending directly translates into specific, measurable reliability improvements, especially for broad reliability programs.
- Re-assess project schedules that appear overly aggressive and may benefit from extended timelines for better design and adaptation to market conditions, such as the E. 179th St. Feeder Establishment.
- Justify the selection of more expensive replacement options (*e.g.*, 750 kVA vs. 500 kVA transformers) based on specific, localized load forecasts rather than broad uprating policies.
- For programs addressing backlogs (*e.g.*, Secondary Open Mains), present a strategy and funding request that aims to control or reduce the total backlog, supported by comprehensive data and risk assessment.

- Review and correct any calculation discrepancies, such as the one identified in the Non-Network Resiliency Program.
- Provide further study or evidence to support the effectiveness of proposed solutions where concerns have been raised, such as for the Selective Undergrounding program's impact on overall outage duration.

By addressing these points, Con Edison can refine its rate filing to ensure that investments are both prudent and timely, aligning expenditures with the most current market realities and implementing a clear, cost-effective strategy for maintaining and enhancing system reliability and resilience for the ratepayers of New York City.

INFORMATION TECHNOLOGY CAPITAL PROJECTS

Q. WHAT ARE YOUR GENERAL OBSERVATIONS REGARDING CON EDISON'S INFORMATION TECHNOLOGY, OR IT, CAPITAL PROJECT PROPOSALS?

A. Con Edison outlines a significant and ambitious plan for technological investments and modernization across various operational and customer-facing domains. Not surprisingly, the Company identifies the need to upgrade aging infrastructure, enhance cybersecurity, improve the customer experience, and leverage data analytics. We recognize the importance of the Company's efforts to address technological obsolescence and prepare for future energy landscape challenges. However, our review of the Company’s testimony and exhibits has identified several recurring themes of concern across multiple proposed IT projects and programs.

Q. WHAT ARE YOUR THEMATIC CONCERNS WITH THE COMPANY'S PROPOSED IT PROJECTS AND PROGRAMS?

A. Our concerns include:

- Premature Funding Requests: Significant capital and O&M funding is often requested before critical planning, scoping, or alternative analysis phases are complete. This practice questions the maturity and reliability of the Company's proposed budgets and timelines, making it difficult to assess the prudence of committing substantial ratepayer funds at this stage.
- Lack of Detailed Cost-Benefit Analyses, or CBA: For many large-scale investments, detailed, quantitative CBAs are notably absent or high-level. The Company's justifications frequently rely on qualitative benefits without providing measurable financial returns for ratepayers (*e.g.*, specific O&M savings, efficiency gains quantified in dollars or FTEs) directly attributable to the proposed expenditures.
- Insufficient Justification for Cost Escalations: Several programs exhibit substantial budget increases, particularly for vendor licensing and ongoing maintenance. These escalations are often attributed to vendor model changes without robust, independent justification, evidence of detailed negotiation efforts, or a thorough evaluation of more cost-effective alternatives.
- Unclear Scope and Cost Allocation: The scope of some multifaceted programs is broad, and the allocation of costs to specific initiatives

within these programs is not always transparent. This lack of granularity makes it difficult to assess the reasonableness of individual components and the overall program budget.

- Potential Strategic Misalignments: In some instances, proposed investments appear to conflict with other stated strategic goals. For example, significant ongoing investment in mainframe technologies is proposed alongside initiatives for mainframe exit, requiring clearer strategic reconciliation. These general concerns underpin many of the specific project critiques that follow and raise questions about the overall prudence and cost-effectiveness of the proposed IT plan from a ratepayer perspective.

Q. WHEN REVIEWING CON EDISON'S PROPOSALS, WHICH PROJECTS DID YOU FIND THAT MAY BE IN NEED OF A REVISED BUDGET FOR 2026 (RY1) OR SUBSEQUENT RATE YEARS?

A. Based on our review of Exhibits IT-1 through IT-12, the following seven significant IT projects and programs may need revised budgets, deferral of certain funding, or require substantial additional justification before their proposed budgets for RY1 and subsequent rate years can be deemed prudent and reasonable:

- Finance and Supply Chain Transformation
- Customer Care & Billing Enhancements and Upgrade
- IT Hardware and Software Maintenance
- Customer Recommendation and Analysis Tools or CARE program

- Billing and Payment Enhancements

Q. PLEASE EXPLAIN WHY EACH PROJECT NEEDS REVIEW.

A. Each of these projects warrants further review due to a combination of factors including the sheer magnitude of the requested funding, the timing of these requests relative to the completion of essential planning phases, the adequacy of the cost-benefit justifications provided, the clarity of the project scope, and the thoroughness of the evaluation of alternatives.

Q. WHAT ARE YOUR CONCERNS REGARDING THE FINANCE AND SUPPLY CHAIN TRANSFORMATION PROJECT?

A. In reviewing Exhibit__(IT-7), we observed that this is an exceptionally large capital undertaking with a total projected cost of $399.8 million. Our primary concern is the prudence of requesting substantial funding for RY1 at $170.8 million and RY2 at $168.5 million *before* the completion of a critical "Phase 0 scoping and planning effort," which is stated to occur in 2025 according to page 60 of Exhibit__(IT-7). This approach makes it difficult to ascertain if the early-year budgets are based on mature and reliable project plans.

Q. DO YOU HAVE OTHER CONCERNS?

A. Yes. For an investment of this scale, the provided white paper lacks a detailed, quantitative cost-benefit analysis that links the $399.8 million expenditure to specific, measurable financial benefits for ratepayers, such as O&M savings or quantifiable efficiency gains. The project aims to move from a heavily customized Oracle EBS system – the Company notes 563 modifications Exhibit__(IT-7) – to a "configuration and not customization" cloud model.

However, a clear strategy and cost implications for managing these existing customizations and the significant business process re-engineering required is not adequately detailed or considered. Finally, with a project conclusion stated for 2032, the full lifecycle costs and the detailed scope of its "Major Transformations" are not transparent enough for the current rate period.

Q. WHAT ARE YOUR RECOMMENDATIONS?

A. We recommend that the Commission defer approval of major RY1 and RY2 implementation funding for the Finance and Supply Chain Transformation project. Initial funding should be limited to amounts demonstrably necessary for the completion and thorough review of the "Phase 0 scoping and planning effort," the development of a robust quantitative CBA, a detailed plan for managing the 563 existing customizations (including business process re-engineering costs), and a full lifecycle cost estimate. Further implementation funding should be strictly contingent upon the Commission's review and approval of these critical deliverables. The Company must also provide detailed breakdowns for the RY1/RY2 "Contract Srvcs" and "Other" capital categories and justify any new FTEs within the capital labor budget.

Q. WHAT ARE YOUR CONCERNS REGARDING THE CUSTOMER CARE & BILLING ENHANCEMENTS AND UPGRADE PROJECT?

A. In reviewing Exhibit__(IT-5), which requests $145.6 million for RY1 through RY3 for this project, we observed that a major concern is the large RY3 budget of $89.4 million allocated for a "CC&B Application Upgrade Implementation." According to Exhibit__(IT-5), the specific upgrade path (on-premise versus

cloud) and its refined costs will only be determined by a "Pre-Planning Project" in 2026. Requesting substantial funding before this critical planning is complete and the solution is defined raises questions, especially as the Company notes on page 77 of Exhibit__(IT-5) that a cloud migration could require *"additional funds beyond those planned for the on-premises solution..."*

The basis for the upgrade cost estimate on page 77 of Exhibit__(IT-5), stated as *"30-40% of the original implementation cost"*, is also unclear as the original implementation cost is not provided. Additionally, the roughly $56.2 million budgeted in RY1 and RY2 for "Ongoing Hardware and Software Enhancements" lacks specific, itemized justifications for the planned initiatives.

Q. WHAT IS YOUR RECOMMENDATION?

A. We recommend deferring approval of the $89.4 million RY3 funding for the "CC&B Application Upgrade Implementation" until the 2026 "Pre-Planning Project" is complete and its findings (including a definitive upgrade path, associated costs, and justification over alternatives) are reviewed by the Commission. Funding for "Ongoing Enhancements" in RY1 and RY2 should only be approved upon receipt of a detailed list of prioritized initiatives with clear individual justifications and cost breakdowns. The Company should also provide the total actual costs of the original CC&B Version 2.8 implementation to provide context for the upgrade estimate.

Q. WHAT ARE YOUR CONCERNS REGARDING THE IT HARDWARE AND SOFTWARE MAINTENANCE PROGRAM)?

A. In reviewing Exhibit__(IT-4), which requests a very significant budget of $299.1 million for RY1 through RY3, we observed that the substantiation for extreme vendor-driven cost increases is insufficient. On pages 48 and 50 of Exhibit__(SSP-4) The Company cites anticipated *"300% increase in Broadcom VMWare licensing cost"* and a similar potential tripling of Broadcom mainframe software licenses based on "market feedback" or past experience. Such extraordinary increases require robust, independent justification, including evidence of negotiation attempts and a thorough evaluation of alternatives, rather than reliance on general statements.

Q. DO YOU HAVE ADDITIONAL CONCERNS?

A. Yes. For such a large O&M program, a detailed cost allocation to specific initiatives (*e.g.*, VMWare licenses, mainframe renewals, cloud services) is necessary but not provided by the Company. There also appears to be a strategic contradiction: the program includes significant O&M for mainframe hardware replacement and software renewals found on page 50 of Exhibit__(SSP-4), which seems at odds with the "Mainframe Exit Data Migration" capital project on page 29 of Exhibit__(IT-8) that suggests a move away from mainframe technology.

Q. WHAT IS YOUR RECOMMENDATION?

A. We recommend that the Commission direct Con Edison to provide detailed, independent substantiation for the anticipated 300% vendor cost increases for VMWare and Broadcom mainframe software, including all vendor communications, market analyses, evidence of negotiation efforts, and a

thorough evaluation of alternatives considered. A granular O&M budget breakdown for RY1 through RY3, itemizing expenditures for each major initiative, is also required.

Furthermore, a clear explanation reconciling significant ongoing mainframe O&M investments with the "Mainframe Exit" strategy is needed. Approval of the full O&M budget, particularly for components with extreme vendor cost increases and ongoing mainframe support, should be contingent on a satisfactory review of this information, with consideration for disallowing unsubstantiated portions based on that outcome.

Q. WHAT ARE YOUR CONCERNS REGARDING THE INFRASTRUCTURE TECHNOLOGY MODERNIZATION PROGRAMS?

A. In reviewing Exhibit__(IT-8), which requests a program budget of $83.73 million and 15 new FTEs, we observed that this project needs further review due to its broad scope and lack of initiative-specific costing in the RY1 request of $34.399 million. The program encompasses diverse initiatives like SRE, IaC, AIOps, and colocation build-outs, and a detailed cost attribution to each is missing for RY1.

The justification for the 15 new FTEs provided on pages 45 to 46 of Exhibit__(IT-8) would benefit from a more robust workload analysis and comparison against contractor alternatives. Moreover, many stated benefits are qualitative, and specific, measurable benefits linked to the costs of individual initiatives are not provided. Finally, potential overlaps with other infrastructure

projects and the rationale for mainframe investments within this program, given a planned mainframe exit, need clarification.

Q. WHAT IS YOUR RECOMMENDATION?

A. We recommend that the Commission direct Con Edison to provide a detailed cost breakdown of the RY1 $34.4 million capital request, allocating funds to each specific initiative (SRE, IaC, AIOps, etc.). A comprehensive workload analysis and justification for each of the 15 new FTEs, including a cost-benefit comparison against contractor options, is needed. The Company must also provide specific, measurable, and quantifiable benefits for each major initiative, linked to costs, and a clear explanation of how this program's data center and mainframe investments are distinct from and do not overlap with other cited capital projects, reconciling mainframe investments with the exit strategy. Full funding of this project should be contingent on this detailed information.

Q. WHAT ARE YOUR CONCERNS REGARDING THE CUSTOMER RECOMMENDATION AND ANALYSIS TOOLS, OR CARE PROGRAM, PROJECT?

A. In reviewing Exhibit__(IT-5), which requests "Strategic" initiative budgeted at $48.2 million for RY1 through RY3, we question the necessity and timing of such a large, complex program and find the Company’s evaluation of alternatives to be lacking. Con Edison dismisses "Alternative #1: Developing small, one-off, home-grown tools" by claiming on page 62 of Exhibit__(IT-5) that it would be *"collectively more costly and less efficient"* but provides no quantitative comparative analysis to support this. Given the recent slowdown in

the development of the clean energy market, the prudence of this large, integrated program versus a more agile, incremental approach using "one-off tools" as specific needs arise would benefit from a more thorough justification.

Q. WHAT DO YOU RECOMMEND?

A. We recommend that the Commission direct Con Edison to provide a detailed comparative analysis, including estimated costs, timelines, functionalities, and risk assessments, for "Alternative #1 (Developing small, one-off, home-grown tools)" versus the proposed integrated CARE program. The Company should also provide a more robust justification for the necessity and timing of the full-scale CARE program, including how its benefits are projected under various scenarios of clean energy market development and customer adoption. The Commission should consider a more phased or modular funding approach based on this information.

Q. WHAT ARE YOUR CONCERNS REGARDING THE BILLING AND PAYMENT ENHANCEMENTS PROJECT?

A. In reviewing Exhibit__(IT-5), which requests a total project budget of $4.0 million for RY through RY3, we have a concern with the justification for advanced kiosk features versus the needs of the target user base, suggesting the inclusion of features that may exceed essential user requirements for the stated target demographic.

The proposal to include features like *"the ability to video conference with a live customer representative"* found on pages 30 to 31 of Exhibit__(IT-5) in kiosks targeting *"unbanked, low-income, and small business customers"*

needs stronger evidence of demand from this specific demographic. The 2023 survey cited by the Company indicated general interest in kiosks as a payment option but does not confirm demand for these advanced, and likely more expensive, interactive features from the intended primary users of this channel, especially when web and mobile platforms are also receiving significant investment.

Q. WHAT DO YOU RECOMMEND?

A. We recommend that the Commission approve funding for the upgrade of kiosks with essential features like broader payment acceptance and language options. However, funding for advanced features such as video conferencing should be deferred or disallowed unless and until Con Edison provides clear evidence of demand from its stated target kiosk users (unbanked, low-income customers) and a justification demonstrating that the incremental benefits of these advanced features outweigh their incremental costs for this specific demographic and channel.

Q. WHAT PROJECTS DO YOU BELIEVE ARE PROPERLY BUDGETED FOR 2026 (RY1)?

A. Based on the information reviewed for the seven significant projects detailed above, we cannot conclude at this time that any of these specific projects are "properly budgeted" for 2026 (RY1) without the substantial additional information and justification requested. Each project has significant flaws related to premature funding requests, lack of detailed cost-benefit analyses, insufficient justification for costs or scope, or unclear alignment with overall

strategy. A determination of proper budgeting can only be made after the Company performs the requested necessary analyses.

Q. BASED ON YOUR REVIEW OF CON EDISON'S IT-RELATED PROPOSALS WHAT ARE YOUR OVERALL RECOMMENDATIONS TO THE COMMISSION?

A. Based on our review, we recommend that the Commission direct Con Edison to:

- Defer Significant Funding Pending Planning Completion: For major transformations like the Finance and Supply Chain Transformation and the CC&B Application Upgrade, defer substantial implementation funding until critical Phase 0/Pre-Planning efforts are complete, reviewed, and approved by the Commission.
- Require Robust Cost-Benefit Analyses: Mandate the provision of detailed, quantitative Cost-Benefit Analyses for all large-scale IT projects, clearly linking investments to measurable financial returns and operational efficiencies, and rigorously evaluating less costly alternatives.
- Demand Substantiation for Cost Escalations: Require thorough, independent substantiation for extreme vendor cost increases, particularly for software licensing (*e.g.*, VMWare, Broadcom mainframe), including evidence of negotiation efforts and evaluation of alternatives, before approving related O&M budgets.

- Ensure Granular Cost Allocation and Scope Clarity: For multifaceted programs like IT Hardware and Software Maintenance and Infrastructure Technology Modernization, require detailed breakdowns of costs allocated to specific initiatives and clear delineation of scope to avoid potential overlaps.
- Reconcile Strategic Misalignments: Direct the Company to provide clear reconciliations where proposed investments (*e.g.*, ongoing mainframe O&M and capital upgrades) appear to conflict with stated strategic goals (*e.g.*, mainframe exit).
- Justify Scope Beyond Baseline Needs: For projects like the "Building Energy Usage Portal" and "Billing and Payment Kiosks," require strong justification and comparative cost analysis when proposed scope and features exceed basic compliance or target user needs.
- Validate Necessity of Large Strategic Programs: For programs like CARE, ensure the necessity and timing of large, integrated solutions are quantitatively justified against more agile, incremental alternatives.
- Strengthen FTE Justifications: Require robust workload analyses and comparisons against contractor cost-effectiveness for all requests for new FTEs associated with IT projects.

By directing the Company to address these points, the Commission can better ensure that IT investments are prudent, cost-effective, and ultimately provide value to ratepayers.

SHARED SERVICES PANEL

Q. WHAT ARE YOUR GENERAL OBSERVATIONS REGARDING CAPITAL PROJECTS ADVANCED BY THE COMPANY'S SHARED SERVICES PANEL TESTIMONY AND EXHIBITS?

A. While Con Edison demonstrated awareness of operational risks and infrastructure gaps across its facilities and support functions in proposing capital projects, the underlying assumptions used to justify several of Con Edison's 2026 capital and O&M budgets rely on future economic and market conditions that may no longer be accurate. There are several capital projects that dominate Con Edison's proposed budget request, relying on strong alignment with clean energy initiatives, operational resilience, and modernization efforts. Con Edison focused on operational efficiency, workforce support, and safety improvements, and also set forth consistent recognition of inflation and market pressures, with annual budget escalations of about 3.5% noted for many projects post-2026.

Q. AFTER REVIEWING THE COMPANY'S MAJOR SHARED SERVICES CAPITAL PROJECTS DO YOU HAVE ANY CONCERNS REGARDING PROJECT BUDGETS?

A. Yes. The assumptions presented by Con Edison in the SSP Testimony and SSP Exhibits 1 through 5 used to justify several of its 2026 capital project budgets rely on future economic and system conditions that do not seem to be entirely accurate. Based on our review of the SSP Exhibits provided by Con Edison, the load growth assumptions appear to reflect an expansionary economy, despite signals that the economy appears to be slowing. Such signals include slowing

and adapting real estate and infrastructure development, stabilizing labor markets, and slowed investment in both public and private infrastructure because of increasing interest rates.

These shifting conditions are not properly reflected in Con Edison's filing for programs tied to somewhat aggressive infrastructure expansion. This is seen through the Third Avenue Garage Infrastructure Improvements Project noted in Exhibit__(SSP-4), where Con Edison is proposing to address operational and safety challenges resulting from the closure of the Third Avenue vehicle repair facility. As noted in Exhibit__(SSP-4) page 93 and Exhibit__(SSP-1) page 1, the Third Avenue Garage Infrastructure Improvements project represents a 5x increase from the 2025 RY1 request of $3 million to $15 million in RY2. Without detailed facility plans or environmental reviews, the feasibility of $15 million in work within a year is not clear.

Q. WHAT ARE YOUR OBSERVATIONS FOR THE SHERMAN CREEK PROJECT?

A. The Sherman Creek project is proposed to be a service center for electric and gas operations employees. The project comprises the development of a new approximately 215,000 square foot service center to support broader urban planning goals, community development, and Con Edison's long-term infrastructure resilience, growth, and efficiency. As noted on page 34 of Exhibit__(SSP-4) and page 1 of Exhibit__(SSP-1), the Company is proposing

a dramatic funding increase from $15 million in 2025 to nearly $71 million in 2026.

This sharp escalation assumes that major construction activities or property acquisitions will proceed without delays within a single rate year. This is an unlikely scenario given the persistent supply chain disruptions, labor constraints, and lengthy permitting processes that typically affect large-scale projects in New York. Con Edison's proposal and supporting workpapers also lacks specifics on permitting status, contractor mobilization, and land acquisition milestones that would support such an aggressive budget increase.

Q. WHAT IS YOUR RECOMMENDATION?

A. Unless Con Edison can provide detailed project management plans, permitting timelines, and scheduling documents to justify the accelerated capital deployment, we recommend that a portion of the proposed budget should be deferred to 2027 to align the budget with a more realistic and industry standard construction timeline.

Q. WHAT ARE YOUR OBSERVATIONS FOR THE THIRD AVENUE GARAGE INFRASTRUCTURE IMPROVEMENT PROJECT?

A. The Third Avenue Garage Infrastructure Improvements project proposes to address major operational and safety challenges resulting from the closure of the Third Avenue vehicle repair facility. As noted on page 93 of Exhibit__(SSP-4) and page 1 of Exhibit__(SSP-1) Con Edison is proposing five times increase from the 2025 request of $3 million to $15 million in 2026. However, Con Edison does not provide any facility engineering plans, site assessments, or

environmental review documents to demonstrate that such a significant amount of work can be scoped, permitted, and executed within a single rate year. It is unclear whether the proposed capital outlay is technically or logistically feasible. Con Edison's justification appears to rest on load growth assumptions that do not fully account for current market signals, specifically a softening in real estate and infrastructure development and a slowing labor market.

Q. WHAT IS YOUR RECOMMENDATION?

A. Given these uncertainties and lack of supporting data from the Company, we recommend that a more cautious and phased funding approach should be taken to ensure that capital investments are timed appropriately and aligned with market conditions.

Q. DOES THIS CONCLUDE YOUR TESTIMONY?

A. Yes, it does.